

test-coverage

LLMOrchestrator — Symbol→Test Coverage Ledger

Round: 275 (deep-doc + Challenge enrichment, mirror round-220 template) **Generated:** 2026-05-19 **Mandate:** CONST-048 (full-automation coverage) + CONST-050 (no-fakes-beyond-unit-tests) + CONST-035 / Article XI §11.9 (anti-bluff)

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17):

"all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Purpose

This ledger maps every exported symbol in the LLMOrchestrator submodule to the test or Challenge that exercises it with runtime evidence. A row without runtime evidence is a §11.4 PASS-bluff regardless of how green the green column reads.

Convention

- **Layer:** unit (mocks allowed per CONST-050(A)), integration (real collaborators, no mocks), challenge (out-of-process bash wrapper, real subprocess + real disk + real wire).

- **Evidence:** anti-bluff artefact emitted during the test (file path, stdout substring, captured JSON). A green PASS column without an evidence column is a critical defect.
- **Mutation:** for every gate, a paired meta-test mutation (§1.1) that inverts the polarity of one invariant. The mutation MUST flip green→red.

Coverage Ledger

Package	Symbol	Layer	Test / Challenge
pkg/parser	DefaultParser.Parse	unit	pkg/parser/ parser_test.go::TestDefault
pkg/parser	DefaultParser.Parse (empty)	unit	pkg/parser/parser_test.go:
pkg/parser	DefaultParser.Parse (5-locale prompts)	challenge	challenges/runner/main.go: llorchestrator_describe_c
pkg/parser	DefaultParser.ExtractJSON	unit	parser_test.go::TestExtract
pkg/parser	DefaultParser.ExtractActions	unit	parser_test.go::TestExtract
pkg/parser	DefaultParser.ExtractIssues	unit	parser_test.go::TestExtract
pkg/protocol	PipeMessage JSON encoding	challenge	challenges/runner/main.go: pipe_test.go::TestPipeMess
pkg/protocol	MessageTypePrompt constant	challenge	challenges/runner/main.go:
pkg/protocol	NewFileTransport	challenge	challenges/runner/main.go: file_test.go::TestNewFileT
pkg/protocol	FileTransport.WriteToInbox	challenge	challenges/runner/main.go: file_test.go::TestWriteToI
pkg/protocol	FileTransport.ReadFromInbox	challenge	challenges/runner/main.go: file_test.go::TestReadFrom
pkg/protocol	FileTransport.WriteSharedFile	unit	file_test.go::TestSharedFi
pkg/protocol	validatePath (private, exercised)	unit+integ	file_test.go::TestPathTrav protocol_integration_test.
pkg/protocol	PipeTransport.Send / Recv	unit	pkg/protocol/pipe_test.go
pkg/i18n	Translator interface	unit	pkg/i18n/ translator_test.go::TestTr
pkg/i18n	NoopTranslator.T	challenge	challenges/runner/main.go:
pkg/i18n	NoopTranslator.TPlural	challenge	challenges/runner/main.go: variant)
pkg/i18n	SetPkgTranslator(nil) + Pkg()		

Package	Symbol	Layer	Test / Challenge
		challenge	challenges/runner/main.go: (Pkg.reset_to_noop)
pkg/agent	Agent interface	unit+integ	pkg/agent/agent_test.go + *_agent_test.go
pkg/agent	AgentPool.Acquire	unit	pkg/agent/pool_test.go::TestAcquire
pkg/agent	SimplePool round-robin	unit	pkg/agent/simple_pool_test.go
pkg/agent	HealthMonitor circuit-breaker	unit	pkg/agent/health_test.go::TestHealthMonitor
pkg/agent	ClaudeCodeAgent.Send	unit	pkg/agent/claudecode_agent_test.go
pkg/agent	GeminiAgent.Send	unit	pkg/agent/gemini_agent_test.go
pkg/agent	JunieAgent.Send	unit	pkg/agent/junie_agent_test.go
pkg/agent	OpenCodeAgent.Send	unit	pkg/agent/opencode_agent_test.go
pkg/agent	QwenCodeAgent.Send	unit	pkg/agent/qwencode_agent_test.go
pkg/agent	MultiPool.Add / Pick	unit	pkg/agent/mock_pool_test.go
pkg/adapter	BaseAdapter lifecycle	unit+integ	pkg/adapter/adapter_test.go + adapter_integration_test.go
pkg/adapter	OpenCodeHeadless.Run	unit	pkg/adapter/opencode_headless_test.go
pkg/config	FromEnv	unit	pkg/config/config_test.go
cmd/orchestrator	main entry	integ	automation_test.go::TestOrchestrator
Chaos	failure injection across pool	challenge	challenges/scripts/chaos_failure_injection_challenge.go
DDoS	health-endpoint flood	challenge	challenges/scripts/ddos_health_flood_challenge.go
Scaling	horizontal pool growth	challenge	challenges/scripts/scaling_horizontal_challenge.go
Stress	sustained-load mix	challenge	challenges/scripts/stress_sustained_load_challenge.go
UI	terminal interaction	challenge	challenges/scripts/ui_terminal_interaction_challenge.go
UX	end-to-end flow	challenge	challenges/scripts/ux_end_to_end_flow_challenge.go

Invariant Floor (per CONST-048)

For every row above, six invariants are asserted:

1. **Anti-bluff posture** — captured runtime evidence per §11.4.
2. **Proof of working capability** — end-to-end on the documented topology.
3. **Implementation matches docs** — README + USER_GUIDE + this ledger reflect actual API.
4. **No open issues** — docs/Issues.md empty for this row or row marked OPERATOR-BLOCKED per §11.4.21.
5. **Documentation in sync** — .md + .html + .pdf mtimes lockstep per §11.4.12 / §11.4.53.
6. **Four-layer test floor** — pre-build + post-build + runtime + paired mutation.

How to Re-validate

```
# Round-275 Challenge (this round's deliverable)
cd dependencies/HelixDevelopment/LLMOrchestrator
bash challenges/llmorchestrator_describe_challenge.sh normal
    # → exit 0
bash challenges/llmorchestrator_describe_challenge.sh mutate
    # → exit 99

# Full unit + integration with race detector
go test -race -count=1 ./...
```